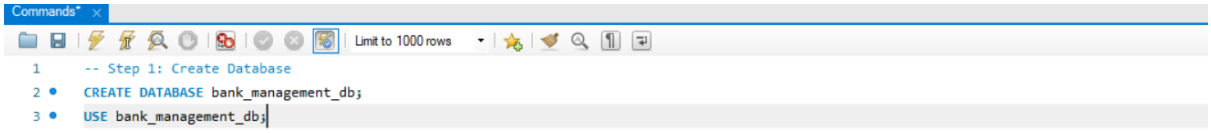
Commands

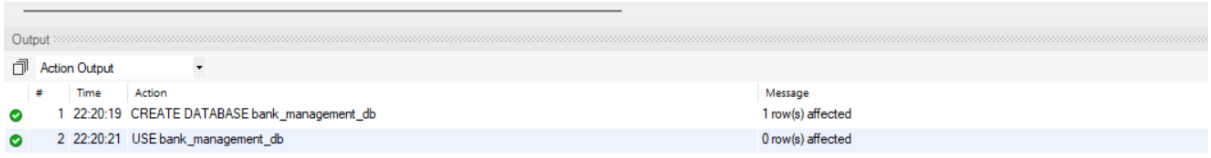
Ili.Harshika (2520030607)

Question :

Perform the following DDL, DML, DCL, and TCL operations on a Bank Management System database: create and alter a bank transaction table, add and modify columns, rename, truncate and drop tables; insert, update, delete, and retrieve transaction records using SELECT, WHERE, and ORDER BY; grant and revoke user privileges; and perform transaction control operations using COMMIT, SAVEPOINT, and ROLLBACK.

Step 1 — Create Database





Step 2 — DDL Commands

Commands*

Limit to 1000 rows

```sql
-- Step 1: Create Database
CREATE DATABASE bank_management_db;
USE bank_management_db;
-- Step 2: Create Bank Transactions Table
CREATE TABLE bank_transactions(
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output

Action Output

#	Time	Action	Message
1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
2	22:20:21	USE bank_management_db	0 row(s) affected
3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected

Commands*

Limit to 1000 rows

```sql
-- Step 1: Create Database
CREATE DATABASE bank_management_db;
USE bank_management_db;
-- Step 2: Create Bank Transactions Table
CREATE TABLE bank_transactions(
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
-- Step 3: Add Account Number Column
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
```

Output

Action Output

#	Time	Action	Message
1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
2	22:20:21	USE bank_management_db	0 row(s) affected
3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Commands

```sql
-- Step 1: Create Database
CREATE DATABASE bank_management_db;
USE bank_management_db;
-- Step 2: Create Bank Transactions Table
CREATE TABLE bank_transactions(
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
-- Step 3: Add Account Number Column
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
-- Step 4: Modify Customer Name Column
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
```

Output

Action Output

#	Time	Action	Message
1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
2	22:20:21	USE bank_management_db	0 row(s) affected
3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Commands

```sql
CREATE DATABASE bank_management_db;
USE bank_management_db;
-- Step 2: Create Bank Transactions Table
CREATE TABLE bank_transactions(
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
-- Step 3: Add Account Number Column
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
-- Step 4: Modify Customer Name Column
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
-- Step 5: Rename Transaction Table
RENAME TABLE bank_transactions
TO customer_transactions;
```

Output

Action Output

#	Time	Action	Message
1	22:20:19	CREATE DATABASE bank_management_db	1 row(s) affected
2	22:20:21	USE bank_management_db	0 row(s) affected
3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected

Commands

Limit to 1000 rows

```
4   -- Step 2: Create Bank Transactions Table
5   CREATE TABLE bank_transactions(
6       txn_id INT PRIMARY KEY,
7       customer_name VARCHAR(50),
8       branch_name VARCHAR(50),
9       transaction_type VARCHAR(20),
10      amount DECIMAL(10,2),
11      transaction_date DATE
12  );
13  -- Step 3: Add Account Number Column
14  ALTER TABLE bank_transactions
15  ADD account_no VARCHAR(20);
16  -- Step 4: Modify Customer Name Column
17  ALTER TABLE bank_transactions
18  MODIFY customer_name VARCHAR(100);
19  -- Step 5: Rename Transaction Table
20  RENAME TABLE bank_transactions
21  TO customer_transactions;
22  -- Step 6: Truncate Transaction Table
23  TRUNCATE TABLE customer_transactions;
```

Output

Action Output

#	Time	Action	Message
2	22:20:21	USE bank_management_db	0 row(s) affected
3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected

Commands

Limit to 1000 rows

```
9       transaction_type VARCHAR(20),
10      amount DECIMAL(10,2),
11      transaction_date DATE
12  );
13  -- Step 3: Add Account Number Column
14  ALTER TABLE bank_transactions
15  ADD account_no VARCHAR(20);
16  -- Step 4: Modify Customer Name Column
17  ALTER TABLE bank_transactions
18  MODIFY customer_name VARCHAR(100);
19  -- Step 5: Rename Transaction Table
20  RENAME TABLE bank_transactions
21  TO customer_transactions;
22  -- Step 6: Truncate Transaction Table
23  TRUNCATE TABLE customer_transactions;
24  -- Step 7: Create Backup Table
25  CREATE TABLE bank_backup(
26      txn_id INT,
27      customer_name VARCHAR(100)
28  );
```

Output

Action Output

#	Time	Action	Message
3	22:21:06	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected
8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected

Commands

Limit to 1000 rows

```
    transaction_date DATE
);
-- Step 3: Add Account Number Column
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
-- Step 4: Modify Customer Name Column
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
-- Step 5: Rename Transaction Table
RENAME TABLE bank_transactions
TO customer_transactions;
-- Step 6: Truncate Transaction Table
TRUNCATE TABLE customer_transactions;
-- Step 7: Create Backup Table
CREATE TABLE bank_backup(
    txn_id INT,
    customer_name VARCHAR(100)
);
-- Step 8: Drop Backup Table
DROP TABLE bank_backup;
```

Output

Action Output

#	Time	Action	Message
4	22:21:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	22:22:07	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected
8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
9	22:24:46	DROP TABLE bank_backup	0 row(s) affected

Step 3: DML Commands

Commands

Limit to 1000 rows

```
CREATE TABLE bank_backup(
    txn_id INT,
    customer_name VARCHAR(100)
);
-- Step 8: Drop Backup Table
DROP TABLE bank_backup;
-- Step 9: Insert Sample Data
INSERT INTO customer_transactions
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
VALUES
(101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Pranitha','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Angela','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Bhanu Priya','Vizag','Deposit',8000,'2024-01-10'),
(105,'Nikshitha','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Manu','Vizag','Deposit',15000,'2024-01-12'),
(107,'Varshika','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Santhvin','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Ravi','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Lahari','Vijayawada','Deposit',11000,'2024-01-16');
```

Output

Action Output

#	Time	Action	Message
6	22:22:54	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected
8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
9	22:24:46	DROP TABLE bank_backup	0 row(s) affected
10	22:27:42	INSERT INTO customer_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'P...	Error Code: 1136. Column count doesn't match value count at row 1
11	22:29:05	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

Commands

```
30  DROP TABLE bank_backup;
31  -- Step 9: Insert Sample Data
32  INSERT INTO customer_transactions
33  (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
34  VALUES
35  (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'),
36  (102,'Pranitha','Hyderabad','Withdrawal',2000,'2024-01-06'),
37  (103,'Angela','Vijayawada','Deposit',12000,'2024-01-08'),
38  (104,'Bhanu Priya','Vizag','Deposit',8000,'2024-01-10'),
39  (105,'Nikshitha','Hyderabad','Withdrawal',3500,'2024-01-11'),
40  (106,'Manu','Vizag','Deposit',15000,'2024-01-12'),
41  (107,'Varshika','Vijayawada','Withdrawal',1000,'2024-01-13'),
42  (108,'Santhvin','Hyderabad','Deposit',9000,'2024-01-14'),
43  (109,'Ravi','Vizag','Withdrawal',4000,'2024-01-15'),
44  (110,'Lahari','Vijayawada','Deposit',11000,'2024-01-16');
45  -- Step 10: Insert New Transaction
46  INSERT INTO customer_transactions
47  (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48  VALUES
49  (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
```

Output

Action Output

#	Time	Action	Message
7	22:23:35	TRUNCATE TABLE customer_transactions	0 row(s) affected
8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
9	22:24:46	DROP TABLE bank_backup	0 row(s) affected
10	22:27:42	INSERT INTO customer_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'P...	Error Code: 1136. Column count doesn't match value count at row 1
11	22:29:05	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	22:31:02	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected

Commands

```
34  VALUES
35  (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'),
36  (102,'Pranitha','Hyderabad','Withdrawal',2000,'2024-01-06'),
37  (103,'Angela','Vijayawada','Deposit',12000,'2024-01-08'),
38  (104,'Bhanu Priya','Vizag','Deposit',8000,'2024-01-10'),
39  (105,'Nikshitha','Hyderabad','Withdrawal',3500,'2024-01-11'),
40  (106,'Manu','Vizag','Deposit',15000,'2024-01-12'),
41  (107,'Varshika','Vijayawada','Withdrawal',1000,'2024-01-13'),
42  (108,'Santhvin','Hyderabad','Deposit',9000,'2024-01-14'),
43  (109,'Ravi','Vizag','Withdrawal',4000,'2024-01-15'),
44  (110,'Lahari','Vijayawada','Deposit',11000,'2024-01-16');
45  -- Step 10: Insert New Transaction
46  INSERT INTO customer_transactions
47  (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48  VALUES
49  (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
50  -- Step 11: Update Transaction Amount
51  UPDATE customer_transactions
52  SET amount = 5000
53  WHERE txn_id = 105;
```

Output

Action Output

#	Time	Action	Message
8	22:24:10	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
9	22:24:46	DROP TABLE bank_backup	0 row(s) affected
10	22:27:42	INSERT INTO customer_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'P...	Error Code: 1136. Column count doesn't match value count at row 1
11	22:29:05	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	22:31:02	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected
13	22:31:41	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Commands

```
(103,'Angela','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Bhanu Priya','Vizag','Deposit',8000,'2024-01-10'),
(105,'Nikshitha','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Manu','Vizag','Deposit',15000,'2024-01-12'),
(107,'Varshika','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Santhvin','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Ravi','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Lahari','Vijayawada','Deposit',11000,'2024-01-16');
-- Step 10: Insert New Transaction
INSERT INTO customer_transactions
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
VALUES
(111,'Venu','Vizag','Deposit',7000,'2024-01-18');
-- Step 11: Update Transaction Amount
UPDATE customer_transactions
SET amount = 5000
WHERE txn_id = 105;
-- Step 12: Delete Transaction
DELETE FROM customer_transactions
WHERE txn_id = 111;
```

Output

Action Output

#	Time	Action	Message
9	22:24:46	DROP TABLE bank_backup	0 row(s) affected
10	22:27:42	INSERT INTO customer_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'P...	Error Code: 1136. Column count doesn't match value count at row 1
11	22:29:05	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	22:31:02	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected
13	22:31:41	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	22:32:15	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected

Commands

```
SET amount = 5000
WHERE txn_id = 105;
-- Step 12: Delete Transaction
DELETE FROM customer_transactions
WHERE txn_id = 111;
-- Step 13: Display All Transactions
SELECT *
FROM customer_transactions;
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Harshika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Nikshitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Varshika	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Ravi	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

transactions 1

Output

Action Output

#	Time	Action	Message
10	22:27:42	INSERT INTO customer_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'...	Error Code: 1136. Column count doesn't match value count at row 1
11	22:29:05	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	22:31:02	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	1 row(s) affected
13	22:31:41	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	22:32:15	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
15	22:32:56	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned

Commands

```
56      WHERE txn_id = 111;
57      -- Step 13: Display All Transactions
58      SELECT *
59      FROM customer_transactions;
60      -- Step 14: Display Deposit Transactions
61      SELECT *
62      FROM customer_transactions
63      WHERE transaction_type = 'Deposit';
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Harshika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

customer_transactions 2

Output

Action Output

#	Time	Action	Message
11	22:29:05	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	22:31:02	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	1 row(s) affected
13	22:31:41	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	22:32:15	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
15	22:32:56	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
16	22:33:30	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned

Commands

```
60      -- Step 14: Display Deposit Transactions
61      SELECT *
62      FROM customer_transactions
63      WHERE transaction_type = 'Deposit';
64      -- Step 15: Sort Transactions by Amount
65      SELECT *
66      FROM customer_transactions
67      ORDER BY amount DESC;
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Harshika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Nikshitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Ravi	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Varshika	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Vijayawada

customer_transactions 3

Output

Action Output

#	Time	Action	Message
12	22:31:02	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	1 row(s) affected
13	22:31:41	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	22:32:15	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
15	22:32:56	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
16	22:33:30	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
17	22:34:28	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

Step 4: DCL Commands

Commands

Limit to 1000 rows

```
55  DELETE FROM customer_transactions
56  WHERE txn_id = 111;
57  -- Step 13: Display All Transactions
58  SELECT *
59  FROM customer_transactions;
60  -- Step 14: Display Deposit Transactions
61  SELECT *
62  FROM customer_transactions
63  WHERE transaction_type = 'Deposit';
64  -- Step 15: Sort Transactions by Amount
65  SELECT *
66  FROM customer_transactions
67  ORDER BY amount DESC;
68  -- Create Auditor1 User
69  CREATE USER 'Auditor1'@'localhost'
70  IDENTIFIED BY '1234';
71
72  -- Create BranchManager User
73  CREATE USER 'BranchManager'@'localhost'
74  IDENTIFIED BY '1234';
```

Output

Action Output

#	Time	Action	Message
13	22:31:41	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	22:32:15	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
15	22:32:56	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
16	22:33:30	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
17	22:34:28	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected

Commands*

Limit to 1000 rows

```
56  WHERE txn_id = 111;
57  -- Step 13: Display All Transactions
58  SELECT *
59  FROM customer_transactions;
60  -- Step 14: Display Deposit Transactions
61  SELECT *
62  FROM customer_transactions
63  WHERE transaction_type = 'Deposit';
64  -- Step 15: Sort Transactions by Amount
65  SELECT *
66  FROM customer_transactions
67  ORDER BY amount DESC;
68  -- Create Auditor1 User
69  CREATE USER 'Auditor1'@'localhost'
70  IDENTIFIED BY '1234';
71
72  -- Create BranchManager User
73  CREATE USER 'BranchManager'@'localhost'
74  IDENTIFIED BY '1234';
75
```

Output

Action Output

#	Time	Action	Message
15	22:32:56	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
16	22:33:30	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
17	22:34:28	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected

Commands

```
59  FROM customer_transactions;
60  -- Step 14: Display Deposit Transactions
61  SELECT *
62  FROM customer_transactions
63  WHERE transaction_type = 'Deposit';
64  -- Step 15: Sort Transactions by Amount
65  SELECT *
66  FROM customer_transactions
67  ORDER BY amount DESC;
68  -- Create Auditor1 User
69  CREATE USER 'Auditor1'@'localhost'
70  IDENTIFIED BY '1234';
71
72  -- Create BranchManager User
73  CREATE USER 'BranchManager'@'localhost'
74  IDENTIFIED BY '1234';
75  -- Step 16: Grant SELECT Permission
76  GRANT SELECT
77  ON customer_transactions
78  TO 'Auditor1'@'localhost';
```

Output

Action Output

#	Time	Action	Message
16	22:33:30	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
17	22:34:28	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected

Commands

```
63  WHERE transaction_type = 'Deposit';
64  -- Step 15: Sort Transactions by Amount
65  SELECT *
66  FROM customer_transactions
67  ORDER BY amount DESC;
68  -- Create Auditor1 User
69  CREATE USER 'Auditor1'@'localhost'
70  IDENTIFIED BY '1234';
71
72  -- Create BranchManager User
73  CREATE USER 'BranchManager'@'localhost'
74  IDENTIFIED BY '1234';
75  -- Step 16: Grant SELECT Permission
76  GRANT SELECT
77  ON customer_transactions
78  TO 'Auditor1'@'localhost';
79  -- Step 17: Grant All Privileges
80  GRANT ALL
81  ON customer_transactions
82  TO 'BranchManager'@'localhost';
```

Output

Action Output

#	Time	Action	Message
17	22:34:28	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected

Commands

```
67  ORDER BY amount DESC;
68  -- Create Auditor1 User
69  CREATE USER 'Auditor1'@'localhost'
70  IDENTIFIED BY '1234';
71
72  -- Create BranchManager User
73  CREATE USER 'BranchManager'@'localhost'
74  IDENTIFIED BY '1234';
75  -- Step 16: Grant SELECT Permission
76  GRANT SELECT
77  ON customer_transactions
78  TO 'Auditor1'@'localhost';
79  -- Step 17: Grant All Privileges
80  GRANT ALL
81  ON customer_transactions
82  TO 'BranchManager'@'localhost';
83  -- Step 18: Revoke SELECT Permission
84  REVOKE SELECT
85  ON customer_transactions
86  FROM 'Auditor1'@'localhost';
```

Output

Action Output

#	Time	Action	Message
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	22:39:55	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected

Commands*

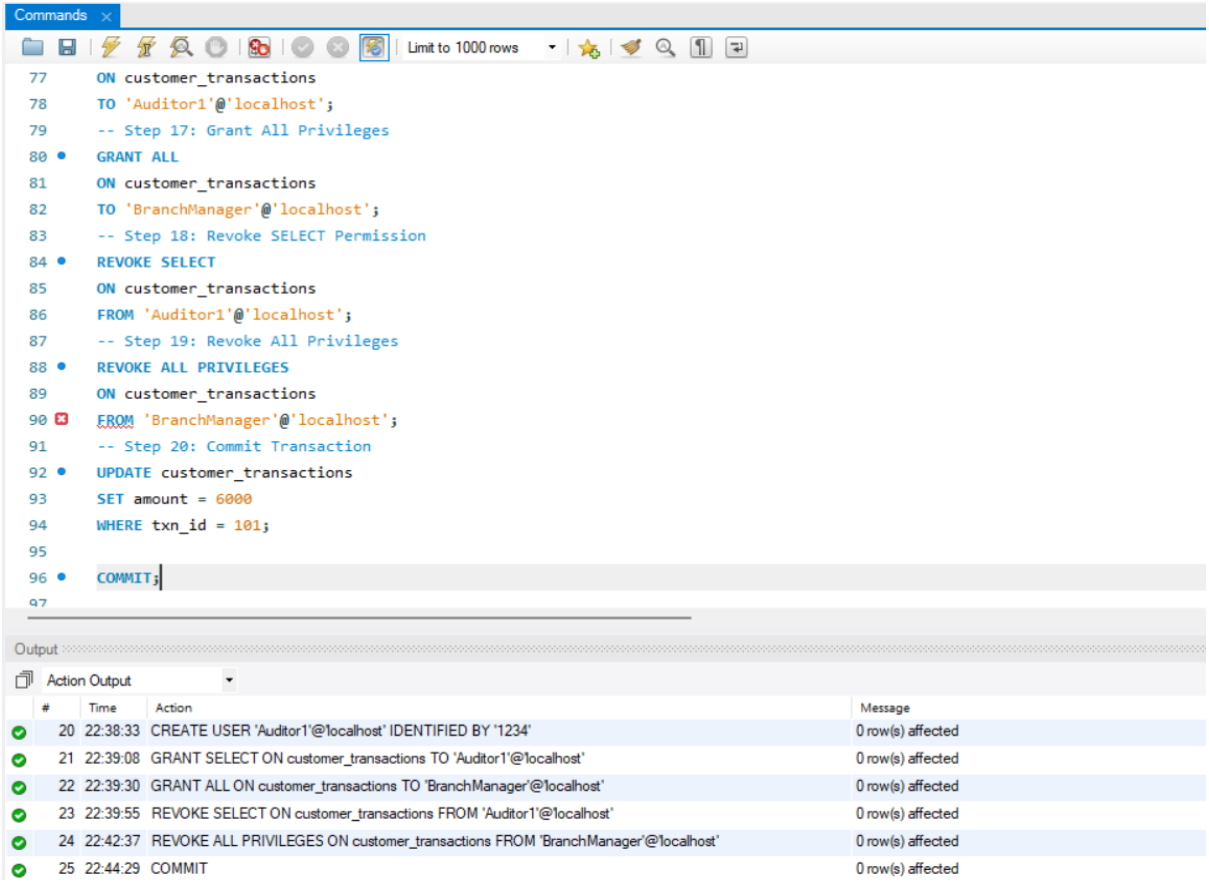
```
72  -- Create BranchManager User
73  CREATE USER 'BranchManager'@'localhost'
74  IDENTIFIED BY '1234';
75  -- Step 16: Grant SELECT Permission
76  GRANT SELECT
77  ON customer_transactions
78  TO 'Auditor1'@'localhost';
79  -- Step 17: Grant All Privileges
80  GRANT ALL
81  ON customer_transactions
82  TO 'BranchManager'@'localhost';
83  -- Step 18: Revoke SELECT Permission
84  REVOKE SELECT
85  ON customer_transactions
86  FROM 'Auditor1'@'localhost';
87  -- Step 19: Revoke All Privileges
88  REVOKE ALL PRIVILEGES
89  ON customer_transactions
90  FROM 'BranchManager'@'localhost';
91
```

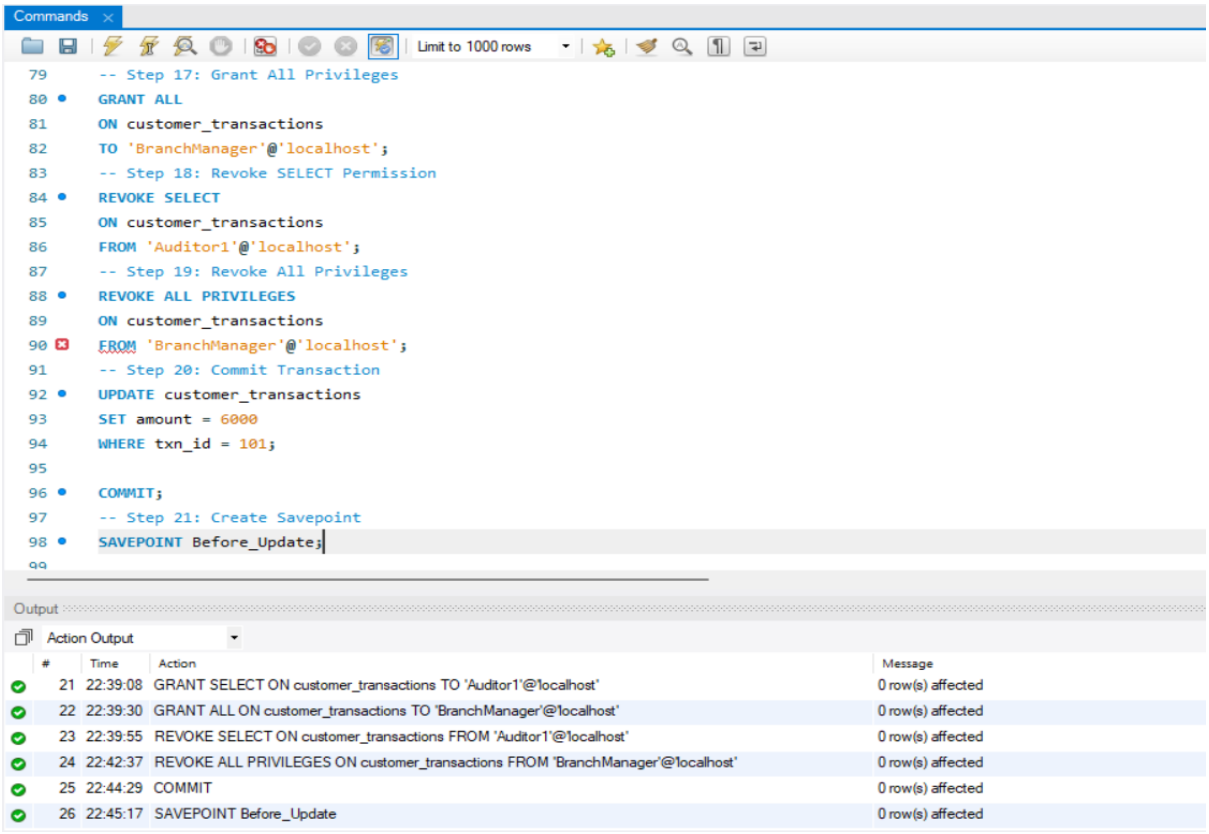
Output

Action Output

#	Time	Action	Message
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	22:39:55	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	22:42:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected

Step 5:TCL Commands





Commands

```
86  FROM 'Auditor1'@'localhost';
87  -- Step 19: Revoke All Privileges
88  REVOKE ALL PRIVILEGES
89  ON customer_transactions
90  FROM 'BranchManager'@'localhost';
91  -- Step 20: Commit Transaction
92  UPDATE customer_transactions
93  SET amount = 6000
94  WHERE txn_id = 101;
95
96  COMMIT;
97  -- Step 21: Create Savepoint
98  START TRANSACTION;
99
100 SAVEPOINT Before_Update;
101 -- Step 22: Rollback to Savepoint
102 UPDATE customer_transactions
103 SET amount = 99999
104 WHERE txn_id = 102;
105
```

Output

Action Output

#	Time	Action	Message
26	22:45:17	SAVEPOINT Before_Update	0 row(s) affected
27	22:45:43	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
28	22:45:56	UPDATE customer_transactions SET amount = 99999 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
29	22:46:04	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
30	22:47:26	SAVEPOINT Before_Update	0 row(s) affected
31	22:47:48	UPDATE customer_transactions SET amount = 99999 WHERE txn_id = 102	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

Commands*

```
109 -- Reset Transaction State
110 ROLLBACK;
111
112 SET autocommit = 1;
113 -- Step 23: Complete TCL Example
114 START TRANSACTION;
115
116 UPDATE customer_transactions
117 SET amount = 7000
118 WHERE txn_id = 101;
119
120 SAVEPOINT SP1;
121
122 UPDATE customer_transactions
123 SET amount = 9000
124 WHERE txn_id = 102;
125
126 ROLLBACK TO SAVEPOINT SP1;
127
128 COMMIT;
129
```

Output

Action Output

#	Time	Action	Message
33	22:50:24	COMMIT	0 row(s) affected
34	22:50:40	SAVEPOINT Before_Update	0 row(s) affected
35	22:51:04	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
36	22:53:20	SET autocommit = 1	0 row(s) affected
37	22:53:48	ROLLBACK TO SAVEPOINT Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
38	22:55:24	COMMIT	0 row(s) affected